

ConSynth-X: Benchmarking Construction CV Under Extreme Field Conditions

Project Progress Report

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Outline

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Problem Statement

Goal: Evaluate how robust construction computer vision models are under extreme field conditions.

Why this matters:

- Construction sites face rain, snow, fog, low light, glare
- Models trained on clean data fail in the field
- No standardized benchmark for these conditions

ConSynth-X provides:

- 1 Synthetic augmentation pipelines (3 methods)
- 2 Multi-label taxonomy of 40+ conditions
- 3 VLM benchmarking framework (11 models)
- 4 YOLOv8 detection benchmarks

6 Condition Categories

Weather (rain, snow, fog)

Lighting (night, glare)

Scale (small/distant)

Image Quality (blur, noise)

Scene Complexity

Domain Shift

Base Datasets

Construction Site Dataset

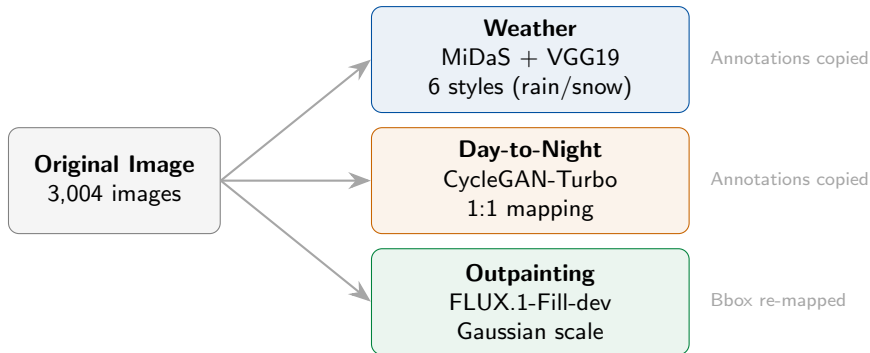
- Format: HuggingFace Arrow
- 3 classes: excavator, rebar, worker_with_white_hard_hat
- Test set: **3,004** images
- Rich annotations: bounding boxes + 4 safety rules + captions

SODA Dataset

- Format: Pascal VOC (XML)
- **15** object classes
- Total: **19,846** images
- Standard VOC train/val/test splits

Condition	Method	Construction Site	SODA
Original	Baseline	3,004	19,846
Weather	Neural Style Transfer	10,266	47,169
Night	img2img-turbo	3,004	–
Small	FLUX Outpainting	1,323	1,001
Total		17,597	68,016

Overview of 3 Augmentation Pipelines

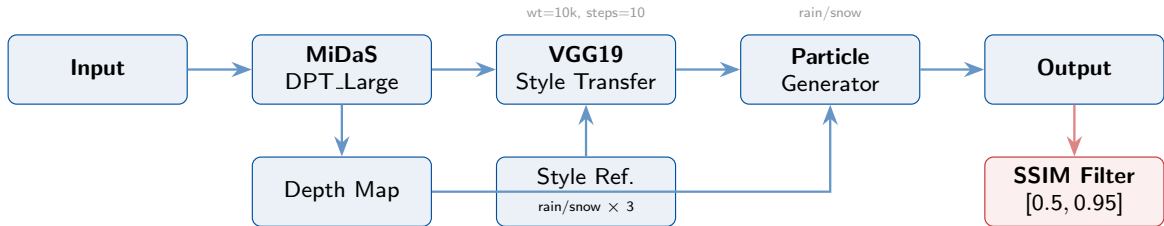


SSIM filter: $0.5 \leq \text{SSIM} \leq 0.95$

512×512 internal, FP16

Scale $\sim \mathcal{N}(0.25, 0.05^2)$, clamp [0.2, 0.4]

Weather Augmentation Pipeline



Construction Site (3,004 original)

SODA (19,846 original)

Style	After Filter	Style	After Filter
rain_0 / rain_1 / rain_2	2,211 / 1,243 / 1,208	rain_0 / rain_1 / rain_2	8,392 / 8,059 / 9,541
snow_0 / snow_1 / snow_2	2,159 / 1,649 / 1,796	snow_0 / snow_1 / snow_2	9,818 / 7,485 / 3,874
Total	10,266	Total	47,169

Day-to-Night Augmentation

Model: CycleGAN-Turbo (day_to_night checkpoint)

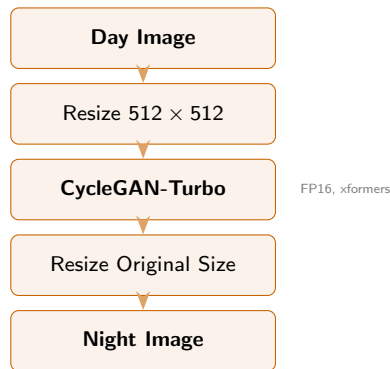
Pipeline:

- 1 Load day image $\rightarrow$ resize to 512×512
- 2 Forward pass through CycleGAN-Turbo
- 3 Resize output back to original resolution (LANCZOS)
- 4 Save night image + copy annotations directly

Key properties:

- **1:1 mapping** – every image gets one night variant
- FP16 inference with xformers attention
- Annotations **copied unchanged** (pixel-level transform)

Output: 3,004 night images (Construction Site)



Outpainting – Scale/Distance Augmentation

Model: FLUX.1-Fill-dev (bfloat16)

Idea: Expand canvas around original image $\rightarrow$ objects appear smaller/more distant.

Scale factor: Truncated Gaussian

$$s \sim \mathcal{N}(\mu=0.25, \sigma=0.05), \quad s \in [0.2, 0.4]$$

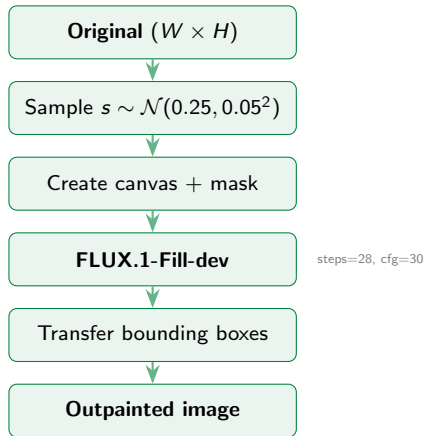
Canvas size: $(1 + 2s) \times$ original on each axis.

Bbox transfer:

$$x'_1 = \frac{x_1 \cdot W \cdot r + p_x}{W'}, \quad y'_1 = \frac{y_1 \cdot H \cdot r + p_y}{H'}$$

where r = resize factor, (p_x, p_y) = paste offset

Prompt: *“an outdoor construction site with buildings, roads, and open sky in the background”*



Dataset	Output
Construction Site	1,323
SODA	1,001

11 Vision-Language Models:

Model	Size
Qwen2.5-VL	7B / 32B / 72B
Qwen3-VL	8B
InternVL2.5	8B / 26B
LLaVA v1.5	7B
Llama-3.2-Vision	11B
Gemma-3	27B
Phi-4-multimodal	–
GPT-4o / GPT-4o-mini	API

4 Benchmark Tasks:

1 **Description** (0-shot captioning)

Metrics: BERTScore, BLEU, ROUGE-L, METEOR, CLIPScore

2 **VQA Safety** (5-shot, 4 rules)

Metrics: Per-rule P/R/F1 + bbox IoU

3 **VQA Rule 1** (binary Yes/No)

PPE compliance detection

4 **Object Detection**

Normalized bbox, grid-based IoU

Conditions: original × weather × night × small

Safety Rules for VQA

Rule 1 – PPE Compliance

- Hard hats, proper clothing, safety shoes
- High-vis vests at night
- Face shields when cutting/welding

Rule 2 – Safety Harness

- Required at height $\geq 3\text{m}$
- Without edge protection

Rule 3 – Edge Protection

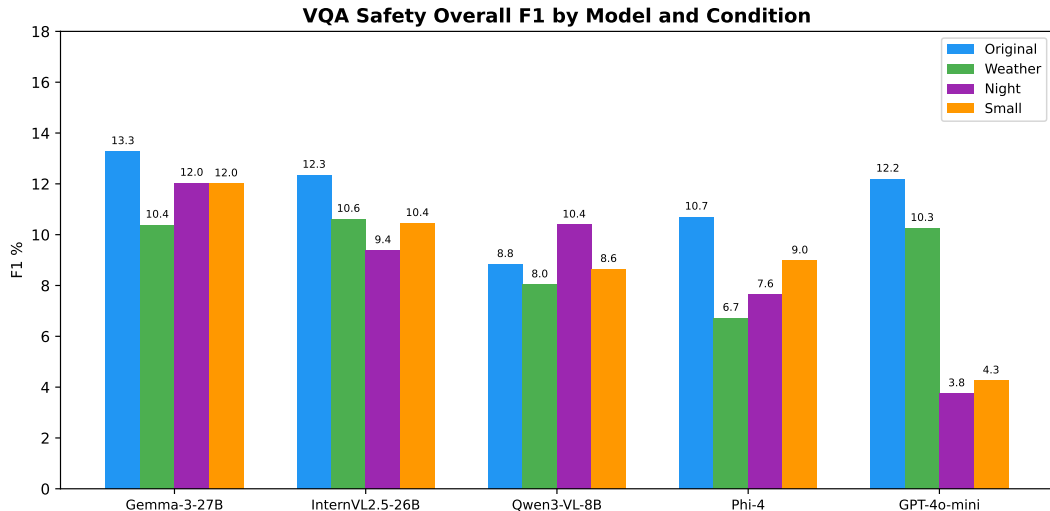
- Guardrails, fences required
- Underground projects $\geq 3\text{m}$ depth

Rule 4 – Excavator Safety

- Workers in blind spots
- Within operation radius

Output format: JSON with rule ID, reason, bounding box $[x_1, y_1, x_2, y_2]$ in normalized $[0, 1]$ space.

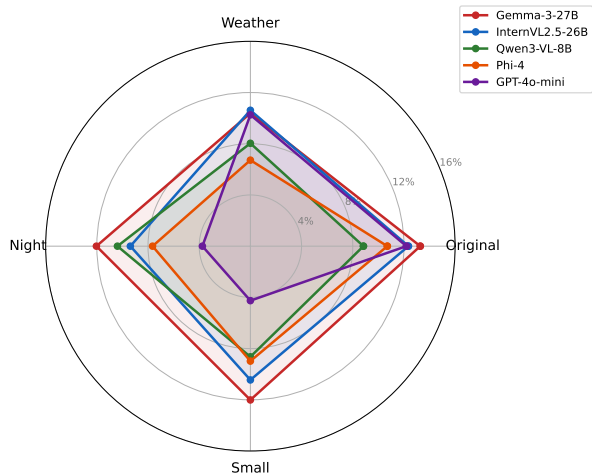
VQA Safety – Overall F1 by Model and Condition



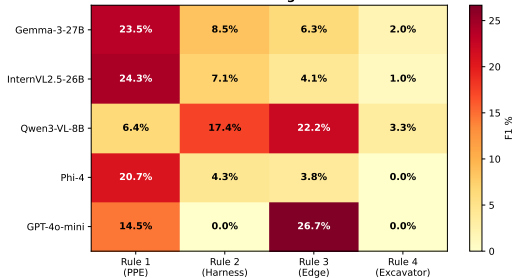
5-shot prompting (Phi-4: 2-shot due to memory). LLaVA-1.5-7B omitted (100% parse errors). Now includes **Gemma-3-27B**.

VQA Safety – F1 Radar Across Conditions

VQA Safety F1 by Condition



Per-Rule F1% on Original Condition



VQA Safety – Key Observations

Condition Impact:

- **Night** and **small** degrade all models
- F1 drops 15–60% from original to worst condition
- **Weather** has moderate impact
- IoU drops more severely than classification metrics

Model Comparison:

- **Gemma-3-27B**: Highest F1 (13.3%), most robust across conditions
- **InternVL2.5-26B**: High recall (77–84%) but low precision
- **Qwen3-VL-8B**: Best bbox IoU (29%), conservative
- **GPT-4o-mini**: ~50% error rate

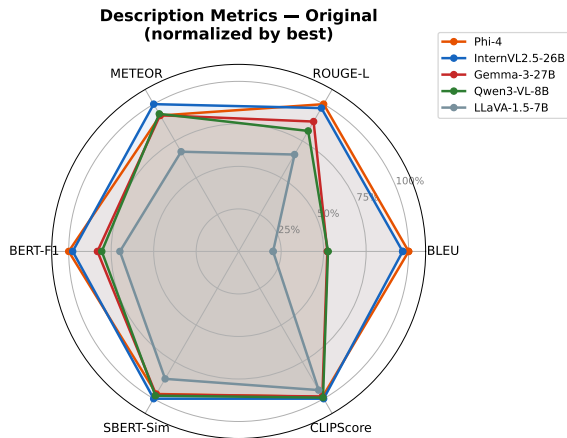
Per-Rule Analysis:

Rule	Best F1	Challenge
Rule 1 (PPE)	0.244	Most common
Rule 2 (Harness)	0.200	Rare violations
Rule 3 (Edge)	0.267	Spatial reasoning
Rule 4 (Excavator)	0.129	Hardest overall

Key finding:

All models struggle with construction safety VQA. Even the best model achieves only ~ 0.30 IoU, suggesting current VLMs lack the spatial and domain understanding needed for reliable safety inspection.

Description Task – Model Profiles (Original)

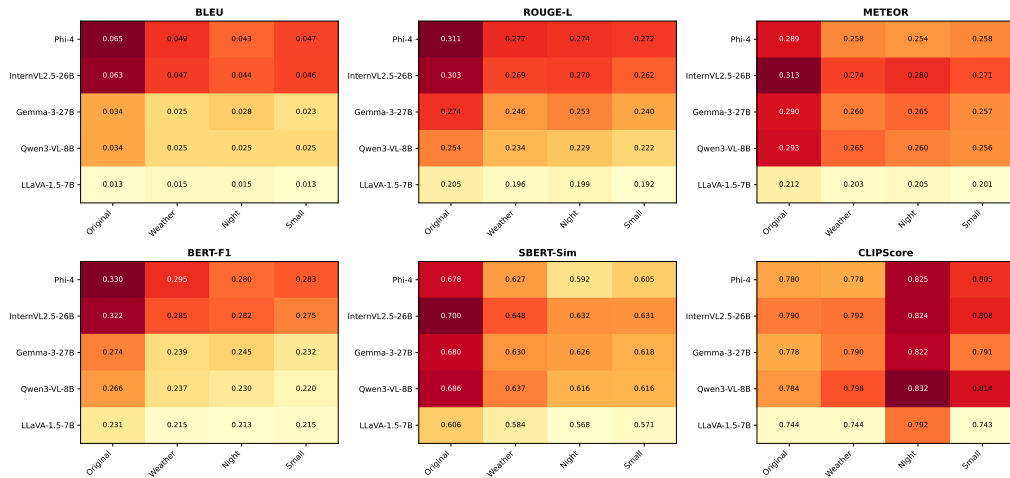


Key findings:

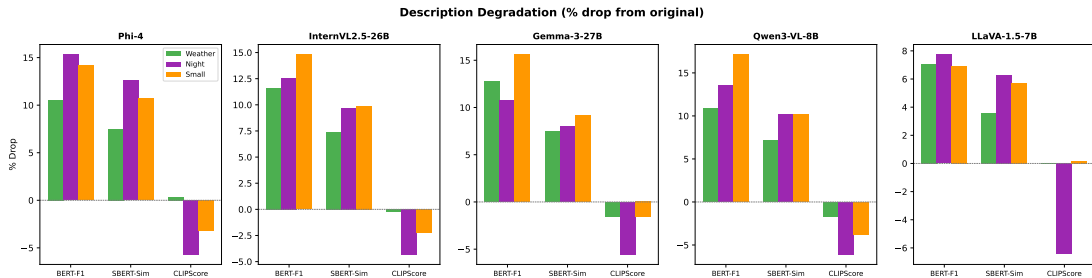
- **Phi-4** and **InternVL2.5-26B** lead on text metrics (BLEU, BERT-F1)
- **CLIPScore** high across all models (0.74–0.83) – descriptions are semantically relevant
- **LLaVA-1.5-7B** weakest across all metrics
- Larger models $\neq$ always better (Phi-4 6B $\approx$ InternVL 26B)

Description Task – All Metrics Heatmap

Description Metrics — All Models x Conditions



Description Task – Degradation Under Augmentation



Positive = performance drop from original. SBERT-Sim and CLIPScore show minimal degradation, confirming augmented images remain semantically consistent.

YOLOv8 Detection – Training Configurations

Construction Site (3 classes):

Config	Training Data
A	Original only
B	Original + Weather
C	Original + Weather + Small
D	Original + Weather + Night
E	Original + Weather + Small + Night

Goal: Measure if augmented data improves robustness across conditions.

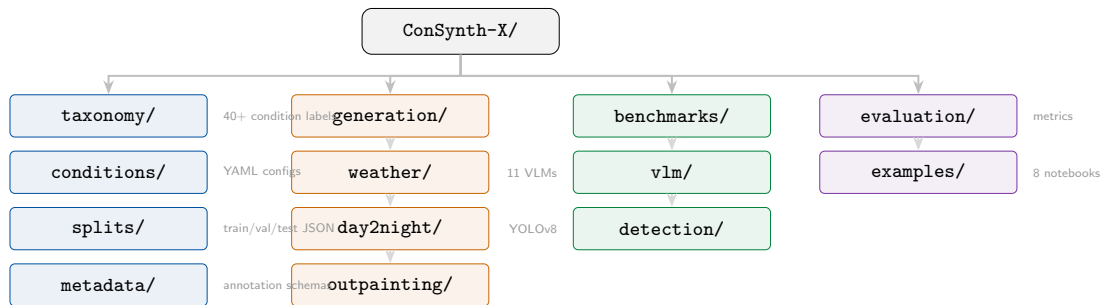
SODA (15 classes):

Config	Training Data
S-A	Original only
S-B	Original + Weather + Night
S-C	Original + Weather + Night + Small

Infrastructure:

- YOLOv8 (nano to xlarge)
- Image size: 640px
- OSC cluster: A100 GPUs
- `fcntl.flock` for parallel SLURM jobs

Repository Structure



Current Status & Next Steps

Completed:

- ✓ 3 augmentation pipelines (weather, night, outpainting)
- ✓ 85,613 total augmented images
- ✓ VLM benchmark framework (11 models)
- ✓ VQA Safety evaluation (5 models $\times$ 4 conditions)
- ✓ YOLOv8 training configs (5+3 experiments)
- ✓ Taxonomy of 40+ conditions
- ✓ Dual format support (Arrow + VOC)
- ✓ Repository consolidated into ConSynth-X

In Progress / Next:

- Run remaining VLMs (Qwen 32B/72B, Gemma-3)
- Description task evaluation (BERTScore, CLIPScore)
- Complete YOLO training & comparison
- Cross-condition evaluation module
- SODA day-to-night augmentation

Planned:

- Per-condition robustness metrics (AP)
- Failure-case visualizations
- Paper with reporting tables

Thank You

Questions?

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